Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2023

Bachelor in Computer Applications

Course Title: **Numerical Methods**

Code No: **CACS252** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

***Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

2. Define error. Explain the Taxonomy Errors.

3. Estimate the value of sin θ at 9 = 45° using Newton's backward difference formula from the following set of data.θ102030405060Sin θ0.17360.34200.50000.64280.76600.8660

4. Write an algorithm and program to calculate integration using Trapezoidal rule.

5. What is the form of resultant matrix using Gauss-Jordan method? Solve the following system of equations using Gauss-Jordan Method.x + 2y - 3z = 42x + 4y - 6z = 8x - 2y - 5z = 4

6. Define ordinary differential equation. Use the fourth order Runge-Kutta method to estimate y(0.4) of the equation:dydx=x2+y2with y(0) = 0 assuming that h = 0.2

7. Solve for the steady state temperatures in a rectangular plate of 8cm x 10cm, if one 10cm side is held at 50°C, and the other 10cm side is held at 30°C and other two sides are held at 10°C. Assume grids of size 2cm x 2cm.

8. Write a short note on (Any Two):a) Partial Differential Equationsb) Linear Interpretationc) Boundary value problems

Group C

Attempt any TWO questions[2x10=20]

9. Write an algorithm and program to compute the root of nonlinear equation using Newton - Raphson method.

10. a) Fit a straight line to the following set of data using Least Square Regression.b) Apply the factorization method (any) to solve the equations:2x + 3y + 7z = 42x + 3y + z = 53x + 4y + z = 7

11. a) Write and implement an algorithm to solve the system of linear equations using Gauss-Seidel method with suitable example.b) Write a program to solve the ordinary differentia; equations using Heun's method.